

Lohitaksha Patary

Computer Science Student | AI, Cloud & Blockchain

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Education

Amrita Vishwa Vidyapeetham, Bengaluru, B.Tech in Computer Science and Engineering – Bengaluru, India Aug 2024 – Present

Experience

AI/Accessibility Intern, IBM SkillsBuild AI Internship – Remote June 2025 – Present

- Built Granite Accessible Assistant — a RAG-powered accessibility kiosk AI using IBM Granite 4.0 Micro, FAISS vector search, and SentenceTransformers, serving users with visual, auditory, and cognitive disabilities
- Designed profile-driven prompt engineering layer adapting responses per disability type and language; added BLIP vision captioning and MediaPipe gesture input for touchless interaction

Cloud Engineer, GDG Amrita Vishwa Vidyapeetham – Bengaluru, India Sept 2024 – Present

- Support study jams and club demos by preparing cloud-native samples and helping peers ship them

Event Organizer, Dastaan Tech Fest 2024 – Bengaluru, India Oct 2024 – Oct 2024

- Coordinated IBM's cloud-native language workshop, covering logistics and on-site execution

Publications

How MOGA Solves the Urban Parking Assignment Problem: Inside the ORCA System 2025

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(DZone)

ORCA — Optimized Routing & Car-park Assignment 2025

Research project applying Multi-Objective Genetic Algorithms on real OSM street networks to assign parking while balancing travel distance, congestion, and sensor cost; benchmarked on Koramangala, Bengaluru. Also available on arXiv.

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PRISM — Probabilistic Real-time Intelligent Scheduling Mechanism 2026

A research simulation comparing traditional OS scheduling algorithms against a hybrid intelligent scheduler using probabilistic pre-analysis (Markov Chains, Poisson arrival rates) and machine learning (Random Forest classification, Weighted Moving Average burst forecasting) to minimize context switching overhead. Also available on arXiv.

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Certifications

Blockchain Opportunity Analysis — INSEAD (Coursera) Feb 2026

8-course professional certification on blockchain strategy and opportunity evaluation under INSEAD and the Blockchain Research Institute. Built PharmaChain — a blockchain-powered medicine supply chain tracker to apply the concepts.

IBM SkillsBuild AI Internship June 2025

Built Granite Accessible Assistant — a RAG-powered accessibility kiosk AI using IBM Granite 4.0 Micro, FAISS vector search, and SentenceTransformers, serving users with visual, auditory, and cognitive disabilities.

Projects

PharmaChain - Blockchain Medicine Tracker Apr 2024 – Present

Blockchain-powered supply chain visibility platform for pharmaceutical batches

- Built Rust backend services exposing REST APIs to a React + TypeScript dashboard
- Added role-aware batch provenance and medicine history tracking for hospitals
- Prototyping Merkle tree verification to enable tamper-evident auditing

IntelliQuery - Natural Language SQL Engine Sept 2024 – Dec 2024

Java service that turns natural-language questions into SQL and explains results for non-technical teams

- Integrated LLM pipeline translating questions into optimized SQL executed on MariaDB via JDBC
- Produced friendly result summaries so HR, admins, and leads can act on data without SQL expertise
- Designed modular architecture enabling conversational access to enterprise datasets

College Canteen Digitization System Feb 2024 – June 2024

Team-led effort to digitize campus canteen ordering with end-to-end web access

- Co-developed ordering, menu, and staff dashboards using Node.js, Express, MongoDB, and campus cloud APIs
- Released a Create React App interface for students to browse, order, and track pickups on any device
- Automated queue updates and payment confirmations to reduce wait times while keeping a clean interface

Campus Carpool Platform July 2024 – Present

Ride-sharing web app for students and faculty built with Next.js, Supabase, and a Create React App client

- Designed Supabase data model with row-level security for rider and driver info
- Added ride pinging, time-based matching, and notifications via Edge Functions and realtime
- Delivered a Create React App interface for campus users with reusable components and a clean experience

Achievements

Global Art Competition

National Level 3rd Runner-up — Regional Art Champion

Edge-Core Hackathon

First Place — Developed an AI-powered touchless hand gesture recognition system using MediaPipe Tasks HandLandmarker with TFLite float16 model bundle, enabling camera-based gesture control for grabbing objects, clicking, and typing without physical contact

Technical Skills

Development: Java, Python, Rust, React, Node.js, Express, Spring Boot, Cloud APIs, Computer Vision, RAG

Platforms & Tools: Ethereum, Solana (Rust), Supabase, Docker, MongoDB, PostgreSQL, FAISS, Figma, AI/ML, Deep Learning, OpenCV, YOLO, MediaPipe, NumPy, Scikit-learn

Extracurricular: Piano Grade 5 Trinity, Music Composition

Languages: English, French, Bengali, Hindi, German